Nikolaos Giakoumidis

AI, Robotics and Automation | Engineering, Research and Technical Leadership

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Explore my projects, laboratories and research: giakoumidis.com

PROFESSIONAL SUMMARY

Engineer, researcher, and technical leader with **15+ years** of experience spanning AI, robotics, electronics, and automation. Led engineering projects, established and managed research laboratories, and brought people, resources, and technologies together to turn ideas into working systems. Combines hands-on technical depth with responsibility for system architecture, integration, procurement, operations, and multidisciplinary collaboration. Working across scientific, technical, institutional, and industry environments has developed a broad perspective and the ability to navigate complex requirements, align diverse stakeholders, and translate unfamiliar problems into practical engineering solutions. Currently leads commercialization and industry engagement at NYU Abu Dhabi's Center for Artificial Intelligence and Robotics, while advancing research in Physical AI through modular architectures that connect agentic reasoning with perception, planning, and coordinated action across heterogeneous robotic systems.

EXPERIENCE

New York University Abu Dhabi (NYUAD) | 2012–Present

Commercial Lead – AI and Robotics | Center for Artificial Intelligence and Robotics (CAIR) | 2025–Present

- Lead commercialization and industry engagement for CAIR, working with faculty, researchers and external partners to identify practical needs and develop applications for AI, robotics and autonomous systems.
- Authored **15+ project proposals** in the first year, translating industrial needs and research capabilities into proposed work across inspection, logistics, infrastructure, digital twins and autonomous systems.
- Completed **five NDAs** with external stakeholders, enabling further technical exchange and collaboration discussions.
- Connect scientific teams with industry and government partners, bringing engineering judgment to discussions of technical feasibility, implementation needs and potential applications.
- Designed and built a **model-agnostic robot agent architecture** for industrial inspection, researching how natural-language task execution can reduce reliance on specialist operators and support practical robotics adoption.
- Represent NYUAD and CAIR through technical demonstrations, executive discussions and exhibitions including ADIPEC, Global Rail, Make it in the Emirates and DriftX.
- Engage organizations across transport, energy, logistics, technology and investment, including AD Ports Group, Etihad Rail, GCAA, Kent, Mubadala, MGX, Analog, CycloTech and Mimik.

Research Instrumentation Specialist – Robotics & Automation | Core Technology Platforms (CTP) | 2017–2025

- Established, expanded and operated laboratories across Robotics, Photonics, High-Throughput Screening, Electronics and Advanced Manufacturing, building shared engineering capabilities for multidisciplinary research.
- Led end-to-end technical decision-making across multiple research laboratories, covering facility design, construction supervision, system architecture, integration and commissioning, equipment strategy, vendor selection and procurement, workflow design, safety procedures, and repair-versus-replacement decisions.
- Supported **100+ users**, including faculty, researchers, students and technical staff, translating experimental requirements into equipment configurations, integrated systems and working research methods.

- Stewarded **US$9M+** in research assets and influenced approximately **US$800K** in annual purchasing, with peak periods approaching **US$3M**, through technical evaluation, vendor selection and procurement planning.
- Diagnosed and restored complex scientific equipment, including systems without vendor support, to maintain research continuity and avoid unnecessary service or replacement expenditure.
- Co-developed and commissioned the Photonics laboratory and managed its operation and expansion. Engineered HTS automation through Thermo Fisher Momentum workflows, FANUC configuration, instrument synchronization and fault recovery.
- Organized approximately four student workshops per semester and co-supervised around two capstone projects annually, mentoring users in engineering methods, system integration and experimental work.
- Operated within demanding safety and regulatory environments involving drones, lasers, and RF systems, while maintaining compliance with university and UAE approval requirements.
- Enabled multidisciplinary research through experimental design, systems integration and scientific instrumentation, with contributions acknowledged in more than 20 publications spanning robotics, photonics, materials science and life sciences.

Electronics Engineer, Laboratories | 2013–2017

- Established the Electronics Workshop and co-established the Advanced Manufacturing Workshop, expanding in-house capabilities in prototyping, scientific instrumentation and engineering support.
- Played a key role in planning and relocating research laboratories from the Center for Science and Engineering to the Saadiyat campus.
- Designed and integrated electronics, control hardware and automation for research teams with different scientific specialisms, supporting experimental development from problem definition through testing.
- Supported the operation and technical development of the High Throughput Screening Platform, integrating robotic automation with scientific instruments and laboratory workflows.
- Delivered in-house troubleshooting and repair of scientific equipment, supporting reliable laboratory operation and informed repair-versus-replacement decisions.
- Presented laboratory capabilities and working systems to industry and government delegations, translating technical research into accessible demonstrations.

Research Assistant | Interactive Robots and Media Laboratory (IRML) | 2012–2013

- Contributed to the establishment and early operation of NYUAD's first robotics laboratory.
- Developed and integrated robotic systems for research, including work linking aerial mapping with ground-robot path planning, and contributed to related publications.
- Served as a teaching assistant for engineering courses and supported international robotics workshops.

Hellenic Armed Forces, Dept. of Advanced Defense Systems | 2011–2012

R&D Engineer | Mandatory national service in Greece

- Conducted R&D on small unmanned aerial systems, including fixed-wing and multirotor platforms.

Experimental Primary School of Athens | 2010–2011

Teacher of Robotics

- Delivered robotics instruction to school students using LEGO Mindstorms, supporting early STEM engagement.

United Arab Emirates University, Al Ain | 2009–2010

Research Intern | Interactive Robots and Media Lab

- Supported robotics research activities and project execution in the Interactive Robots and Media Lab.

G&D Goumas SA, Athens | 2003–2009

Manager of Industrial and Automotive Equipment

- Managed technical support and training activities for industrial and automotive diagnostic equipment, including Launch diagnostic tools and troubleshooting systems.

EDUCATION

University of the Aegean, Greece

PhD Candidate | Started 2023 | Expected completion: January 2027
Thesis: Modular Architectures for Physical AI: Agentic Reasoning and Orchestration in Heterogeneous Robotic Systems.

University of West Attica, Greece

BSc, Automation Engineering | 2012

SELECTED CERTIFICATIONS

- Dubai Civil Aviation Authority | Professional RPAS Certificate, Multirotor & Fixed Wing, 2023
- NVIDIA Deep Learning Institute | Deep Learning for Robotics, 2019
- KUKA College | KUKA LBR iiwa Commissioning and Programming, 2019
- Udacity | Deep Reinforcement Learning Nanodegree, 2019

AWARDS

- Team First Prize, RTA Dubai World Challenge for Self-Driving Transport | Delivery Drone, 2021
- First Prize, Design EmbedIT Competition, Shanghai, 2012
- First Prize, World Robot Olympiad, Greek national round, 2009
- First Prize, Digital Week Robotics Competition, Athens, 2008

CORE COMPETENCIES

LLMs, VLMs and Agentic AI

Model-agnostic agent architectures and MCP skill servers connecting LLM/VLM reasoning with software tools, robot sensing and actions. Multi-step tool use, structured outputs, persistent memory and sub-agent delegation; multimodal image, video and speech pipelines. Retrieval-augmented generation, hybrid lexical/semantic retrieval, embeddings and entity-relation retrieval. VLM fine-tuning with LoRA and GRPO using PyTorch, Hugging Face Transformers and PEFT; DGX Spark for model development and fine-tuning. Agent orchestration with n8n and LangGraph; durable workflows with Temporal, deterministic execution constraints and human approval controls.

Programming, Edge AI Deployment and Evaluation

Python, C++, C, TypeScript, MATLAB and LabVIEW; FastAPI, REST APIs, PostgreSQL and Docker. More than a decade of hands-on experience with NVIDIA technologies, including onboard AI deployment on robots and drones across five Jetson generations, from TX1 to Thor. TensorRT optimization of classification models and LLMs; CUDA-accelerated processing, local model serving with Ollama and asynchronous service integration. Comparative model evaluation, structured-output validation, accuracy/latency assessment, execution tracing and failure analysis. Evidence provenance, idempotency and recovery mechanisms; supporting experience with TensorFlow and LangChain.

Cooperative Robotics, Reinforcement Learning and Simulation

Heterogeneous multi-robot systems, cooperative exploration, autonomous inspection and data collection. Doctoral research includes training Spot locomotion policies for varied terrain representative of archaeological environments and sim-to-real deployment on the physical robot. Isaac Lab workflows covering custom terrain generation and curricula, height-scan and camera-based policies, PPO training, RSL-RL, rl_games, Stable-Baselines3 and TensorBoard. Isaac Sim for realistic robotics simulation, synthetic data generation and algorithm evaluation; ArduPilot SITL integration, Gazebo and MATLAB Simulink. Localization, mapping, navigation, path planning and next-best-view methodologies; human–robot interaction, shared autonomy and teleoperation. ROS 2, robot SDKs and MAVLink; broad familiarity with Isaac ROS GPU-accelerated packages for perception, localization, mapping and robot autonomy. Route and constrained optimization with cuOpt and Gurobi, including comparative benchmarking.

Computer Vision and Multimodal Perception

Integration and experimental validation of deep-learning perception systems, including UAV detection, visual tracking, RGB–thermal sensing and cooperative visual localization. Research contributions to Siamese transformer-based aerial tracking, pan-tilt-zoom tracking and spherical-camera localization. YOLO-based detection integrated with ROS 2; NVIDIA Cosmos for image and video understanding in robot agent workflows. LiDAR and RGB-D processing, TSDF fusion, point-cloud reconstruction and camera-pose transformations. Real-to-sim scene generation from 3D scans and RGB-D data using Open3D and OpenUSD.

Robotic Platforms and Hardware Integration

Ground, aerial and underwater robotics; Boston Dynamics Spot, Clearpath Husky, Unitree H1/G1, KUKA, FANUC, Kinova and ST Robotics; PX4, ArduPilot and DJI ecosystems. Electronics, mechatronic integration, embedded interfaces, laboratory automation and scientific instrumentation. LiDAR, radar, RGB/thermal, multispectral and acoustic imaging, motion capture, GNSS, IMUs and 3D scanners; optical/RF systems, SCPI instruments and serial telemetry.

Engineering Leadership and Research Translation

System architecture, requirements definition, technical assessment, experimental development and field validation. Laboratory establishment, systems integration, commissioning, equipment strategy, procurement, vendor coordination, resource planning, safety procedures and fault diagnosis. Industry needs discovery, proposal development, technical demonstrations and stakeholder communication; multidisciplinary collaboration, mentoring and user training.

SELECTED PUBLICATIONS

- Giakoumidis, N., Tzes, A., Anagnostopoulos, C.-N. *“A Zero-Training Robot Agent Harness for Natural-Language Field Inspection.”* **Manuscript under review**, Journal of Field Robotics, 2026.
- Azim, A. W., Bazzi, A., Bomfin, R., Giakoumidis, N., Rappaport, T. S., Chafii, M. *“Indoor Statistical and Deterministic RCS Characterization for ISAC Channel Modeling.”* IEEE Transactions on Wireless Communications, 25, 21337–21354, 2026.
- Giakoumidis, N., Anagnostopoulos, C.-N. *“ARM4CH: A Methodology for Autonomous Reality Modelling for Cultural Heritage.”* Sensors, 24(15), 4950, 2024.
- Prieto, S. A., Giakoumidis, N., García de Soto, B. *“Multiagent Robotic Systems and Exploration Algorithms: Applications for Data Collection in Construction Sites.”* Journal of Field Robotics, 41(4), 1187–1203, 2024.
- Xing, D., Tsoukalas, A., Evangeliou, N., Giakoumidis, N., Tzes, A. *“Siamese Adaptive Transformer Network for Real-Time Aerial Tracking.”* ICUAS, 2022.
- Xing, D., Tsoukalas, A., Giakoumidis, N., Tzes, A. *“Computationally Efficient RGB-t UAV Detection and Tracking System.”* ICUAS, 1410–1415, 2021.
- Eid, M. A., Giakoumidis, N., El Saddik, A. *“A Novel Eye-Gaze-Controlled Wheelchair System for Navigating Unknown Environments: Case Study With a Person With ALS.”* IEEE Access, 4, 558–573, 2016.

Research record: **31 listed research outputs** | **500+ citations**.
Full publication list and citation metrics: Google Scholar
https://scholar.google.com/citations?user=HmOOogwAAAAJ&hl=en